

Lecture 2

Some examples of probability converge

WLLN

s-7 example $\rightarrow$ almost sure convergence
SLLN

Thm for equivalent statement of SLLN
a.s. convergence

Corollary

e.g. $Y \sim \text{Unif}[0, 5]$ $X_n = (1 + \frac{1}{n})Y$ $X_n \xrightarrow{P} Y$?

观察 here $|X_n - Y| = |(1 + \frac{1}{n})Y - Y| = |\frac{1}{n}Y| = \frac{1}{n}Y \leq \frac{5}{n}$

$\therefore$ 当 $\frac{5}{n} < \varepsilon$, $|X_n - Y| \leq \frac{5}{n} < \varepsilon$

$\therefore$ we can take $n > \frac{5}{\varepsilon}$

Pf: If $\varepsilon > 0$, $\forall n > \frac{5}{\varepsilon}$, $P(|X_n - Y| \geq \varepsilon) = 0$

$\therefore \lim_{n \rightarrow \infty} P(|X_n - Y| \geq \varepsilon) = 0 \Rightarrow X_n \xrightarrow{P} Y$

e.g. (312 WLLN)

Flip many fair coins

$X_n = \begin{cases} 1 & \text{nth coin heads} \\ 0 & \text{nth coin tails} \end{cases}$

Does $X_n \rightarrow \frac{1}{2}$?

Answer no!

If $0 < \varepsilon < \frac{1}{2}$, $P(|X_n - \frac{1}{2}| \geq \varepsilon) = P(X_n = 1) + P(X_n = 0) = 1$ $\rightarrow 0$

What about

$M_n = \frac{1}{n}(X_1 + X_2 + \dots + X_n)$ = fraction of heads in first n flips

Lecture 2

3/2 ↓ Weak Law of Large Numbers (WLLN)

Thm: If $X_1, X_2, \dots$ indep, each with finite mean μ ,
variance $\leq v < \infty$

then, if $M_n = \frac{1}{n}(X_1 + \dots + X_n)$

$$M_n \xrightarrow{P} \mu$$

Pf of the Thm:

$$E[M_n] = \frac{1}{n}(\mu + \mu + \dots + \mu) = \frac{1}{n} \cdot n\mu = \mu$$

$$\text{Var}(M_n) \leq \frac{1}{n^2}(v + v + \dots + v) = \frac{1}{n}v$$

So, by Chebyshev's inequality,

$$P(|M_n - \mu| \geq \varepsilon) \leq \frac{\text{Var}(M_n)}{\varepsilon^2} \leq \frac{v}{n\varepsilon^2} \rightarrow 0$$

So $M_n \xrightarrow{P} \mu$ //

Fact of Extension:

If $\{X_n\}$ are iid. with finite mean μ , then WLLN holds even if $v = \infty$.

5-7 example (3/2 ~~SLN~~)
almost sure convergence

e.g. Define $X_1, X_2, \dots$ as follows. Most $X_n = 5$.

But, one of $X_1, \dots, X_9 = 7$ chosen unif at random

And, one of $X_{10}, \dots, X_{99} = 7$ chosen unif at random

one of $X_{100}, \dots, X_{999} = 7$

one of $X_{1000}, \dots, X_{9999} = 7$

$$(X_n) = \underbrace{5555 \ 7 \ 5555}_{1-9} \underbrace{555 \dots 5 \ 7 \ 5 \dots 5}_{10-99} \underbrace{5555 \dots 5 \ 7 \ 5 \dots 5}_{100-999} \dots$$

Then, does $X_n \xrightarrow{P} 5$?

Answer Yes!

$$P(X_n = 7) = \begin{cases} \frac{1}{9} & 1 \leq n \leq 9 \\ \frac{1}{90} & 10 \leq n \leq 99 \\ \frac{1}{900} & 100 \leq n \leq 999 \\ \vdots & \vdots \end{cases} = \frac{1}{9 \times 10^{k-1}} \quad \text{when } n \text{ is a } k \text{ digit number} \\ \text{(n has } k \text{ digits)}$$

$$\text{then } \lim_{n \rightarrow \infty} P(X_n = 7) = 0$$

$$\text{So, } \lim_{n \rightarrow \infty} P(|X_n - 5| \geq \epsilon) \leq \lim_{n \rightarrow \infty} P(X_n = 7) = 0$$

$$\text{So, } X_n \xrightarrow{P} 5$$

However, $X_n \xrightarrow{a.s.} 5$ what does it mean?

Defn. Almost Sure Convergence

A sequence of R.V. $X_1, \dots, X_n$ converges to γ almost surely or with probability one or w.p.1 if

$$P(X_n \rightarrow \gamma \text{ as a sequence of real numbers}) = 1$$

$$X_n \xrightarrow{a.s.} \gamma \text{ denoted}$$

通过 5-7 example, 能感受到 almost sure convergence 更强一点.

$$\therefore \text{Thm } X_n \xrightarrow{a.s.} \gamma \Rightarrow X_n \xrightarrow{P} \gamma$$

Thm cts ↓

Thm: If $X_n \xrightarrow{a.s.} Y$, then $X_n \xrightarrow{P} Y$

Pf: Fix $\varepsilon > 0$

$$\text{Let } A_n = \{\exists m \geq n \text{ s.t. } |X_m - Y| \geq \varepsilon\} \\ = \bigcup_{m=n}^{\infty} \{|X_m - Y| \geq \varepsilon\}$$

$$\text{Then } \bigcap_{n=1}^{\infty} A_n = \{|X_n - Y| \geq \varepsilon \text{ i.o.}\} \subseteq \{X_n \not\rightarrow Y\}$$

$$\bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} \{|X_m - Y| \geq \varepsilon\}$$

$$\text{Since } X_n \xrightarrow{a.s.} Y, P(X_n \not\rightarrow Y) = 0.$$

$$\text{So, } P\left(\bigcap_{n=1}^{\infty} A_n\right) = 0$$

$$A_n = \bigcup_{m=n}^{\infty} \{|X_m - Y| \geq \varepsilon\} \Rightarrow A_n \text{ is decreasing, } A_{n+1} \subseteq A_n, \\ \text{i.e. } \{A_n\} \searrow \bigcap_{n=1}^{\infty} A_n$$

So, By continuity of probability

$$\lim_{n \rightarrow \infty} P(A_n) = P\left(\bigcap_{n=1}^{\infty} A_n\right) = 0$$

$$A_n \supseteq \{|X_n - Y| \geq \varepsilon\}$$

$$\therefore \lim_{n \rightarrow \infty} P(|X_n - Y| \geq \varepsilon) \leq \lim_{n \rightarrow \infty} P(A_n) = 0 \quad \forall \varepsilon$$

$$\therefore X_n \xrightarrow{P} Y \quad \square$$

- do almost sure convergence is example.

e.g. $U \sim \text{Unif}[0,1]$

$$X_n = \mathbb{1}_{U \leq \frac{1}{2} + \frac{1}{2^n}} \quad Y = \mathbb{1}_{U \leq \frac{1}{2}} \quad \text{know } X_n \xrightarrow{P} Y$$

Does $X_n \xrightarrow{\text{a.s.}} Y$?

PS: (解决方法 — 分类讨论 + 观察)

Case one: If $U \leq \frac{1}{2}$, $Y = 1$

$$X_n = 1 \quad \forall n \Rightarrow X_n \rightarrow Y$$

Case two: If $U > \frac{1}{2}$, $Y = 0$.

$$\text{recall } X_n = \begin{cases} 1 & U \leq \frac{1}{2} + \frac{1}{2^n} \\ 0 & \text{otherwise} \end{cases}$$

$\therefore$ when $\frac{1}{2^n} < U - \frac{1}{2}$, we have $U > \frac{1}{2} + \frac{1}{2^n}$, then $X_n = 0$

$$\begin{aligned} &\nearrow \\ 2^{-n} &\therefore \text{when } \frac{1}{2^n} < U - \frac{1}{2} \Leftrightarrow n > \log_2\left(\frac{1}{U - \frac{1}{2}}\right), \\ &X_n = 0 \end{aligned}$$

$$\text{and } X_n \rightarrow Y$$

So $X_n \xrightarrow{\text{a.s.}} Y$

Still the same example, but 一点巧思.

e.g. Suppose instead

$$U \sim \text{Unif}[0,1] \quad X_n = \mathbb{1}_{U \leq \frac{1}{2} + \frac{1}{2^n}} \quad Y = \mathbb{1}_{U < \frac{1}{2}}$$

case 1: $U < \frac{1}{2}$ $X_n \rightarrow Y$

case 2: $U > \frac{1}{2}$ $X_n \rightarrow Y$

case 3: $U = \frac{1}{2}$ $Y = 0$ but $X_n = 1 \quad \forall n$

$$\text{So, } X_n \rightarrow Y \text{ iff } U \neq \frac{1}{2}$$

However, $X_n \xrightarrow{\text{a.s.}} Y$ as before

continuous pdf 单点概率为 0!

$$\text{Reason: } P(X_n \rightarrow Y) = P(U \neq \frac{1}{2}) = 1 - P(U = \frac{1}{2}) = 1 - 0 = 1$$

$$X_n \xrightarrow{\text{a.s.}} Y$$

e.g. $\{Z_n\}$ indep, $Z_n \sim \text{Bernoulli}(\frac{1}{n})$

$$\text{i.e. } \begin{cases} P(Z_n=1) = \frac{1}{n} \\ P(Z_n=0) = 1 - \frac{1}{n} \end{cases}$$

① ~~XXXXXX~~ $Z_n \xrightarrow{P} 0$?

Yes!

$$\forall \epsilon > 0, P(|Z_n - 0| \geq \epsilon) \leq P(Z_n = 1) = \frac{1}{n} \rightarrow 0$$

$$\text{So } Z_n \xrightarrow{P} 0$$

② $Z_n \xrightarrow{\text{a.s.}} 0$?

No!

$$\sum_{n=1}^{\infty} P(Z_n=1) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty \quad \text{and } \{Z_n\} \text{ indep}$$

$$\Rightarrow \text{By Borel-Cantelli, } P(Z_n=1 \text{ i.o.}) = 1$$

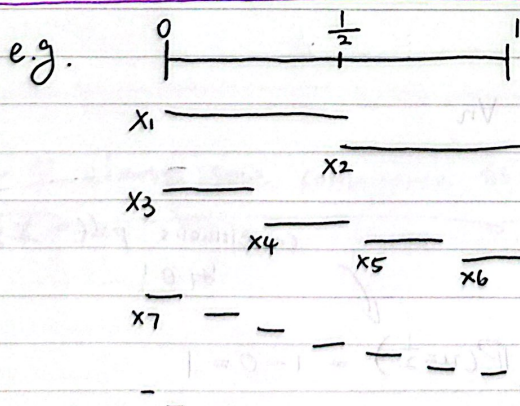
$$\text{So } P(Z_n \rightarrow 0) = 0 \quad \text{So } Z_n \not\xrightarrow{\text{a.s.}} 0$$

怎么改成 $Z_n \xrightarrow{\text{a.s.}} 0$

If, $Z_n \sim \text{Bern}(c)$ $c = \frac{1}{n^2}, 2^{-n}, \text{etc.}$

$$\text{then } \sum_{n=1}^{\infty} P(Z_n=1) < \infty \Rightarrow \text{By Borel-Cantelli, } P(Z_n=1 \text{ i.o.}) = 0$$

$$\therefore Z_n \xrightarrow{\text{a.s.}} 0$$



$$U \sim \text{Unif}[0, 1]$$

$$X_1 = \mathbb{1}_{U \in [0, \frac{1}{2}]}$$

$$X_2 = \mathbb{1}_{U \in (\frac{1}{2}, 1]}$$

问题: Does $X_n \xrightarrow{\text{a.s.}} 0$ in prob? a.s.?

思路: $X_n \xrightarrow{P} 0$ 因为随着每个 X 的区间越小, $X=1$ 的概率越小

$X_n \xrightarrow{a.s.} 0$ 每次循环之间就会有一组 X 覆盖区间, 导致 $P(X_n=1 \text{ i.o.})=1$

But how to prove?

Solution: 找出可以表达概率的方式

本身 $X_1, X_2, \dots, X_n$ 无法定位具体区间

Switch to K : 第几组 j : $(0, 1, 2, \dots, 2^k-1)$
(一个组有 2^k 个元素) 组里的第几个

终于, 可以转化

$$X_1 = \mathbb{1}\{u \in [0, \frac{1}{2}]\} \quad X_2 = \mathbb{1}\{u \in [\frac{1}{2}, 1]\}$$

$$X_3 = \mathbb{1}\{u \in [0, \frac{1}{4}]\} \quad X_4 = \mathbb{1}\{u \in [\frac{1}{4}, \frac{1}{2}]\} \quad X_5 = \mathbb{1}\{u \in [\frac{1}{2}, \frac{3}{4}]\} \quad X_6 = \mathbb{1}\{u \in [\frac{3}{4}, 1]\}$$

switch to

$$\Rightarrow X_{2^k-1+j} = \mathbb{1}\{u \in [\frac{j}{2^k}, \frac{j+1}{2^k}]\}$$

(For each)

$k=1, 2, \dots$

$j=0, 1, 2, \dots, 2^k-1$

$$\Rightarrow \begin{array}{c} |X_1 X_2| \quad |X_3 X_4 X_5 X_6| \quad |X_7 X_8 \dots X_{14}| \dots \\ \uparrow \quad \quad \uparrow \quad \quad \uparrow \\ k=1 \quad k=2 \quad k=3 \end{array}$$

Prove: $X_n \xrightarrow{P} 0$

$$\text{Let } \varepsilon > 0, \quad P(|X_n - 0| \geq \varepsilon) = P(|X_n| \geq \varepsilon)$$

$$\leq P(X_n = 1)$$

$$= P(X_{2^k-1+j} = 1) = P(u \in [\frac{j}{2^k}, \frac{j+1}{2^k}))$$

$$= \frac{1}{2^k} \rightarrow 0$$

$$\therefore X_n \xrightarrow{P} 0$$

as $n \rightarrow \infty$

($k \rightarrow \infty$)

continuous
proving

prove:

WTS $X_n \xrightarrow{a.s.} 0$

$$P(X_n \rightarrow 0) \neq 1$$

Remember: we switch to X_n to $X_{2^k-1+j} = \mathbb{1}_{u \in [\frac{j}{2^k}, \frac{j+1}{2^k})}$

First, we define $A_n := \{X_n = 1\}$

$$P(A_n \text{ i.o.}) = P\left(\bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n\right) = \lim_{N \rightarrow \infty} P\left(\bigcup_{n=N}^{\infty} A_n\right)$$

↑
By continuity of probability

Let $N > 0$, $P\left(\bigcup_{n=N}^{\infty} A_n\right)$

choose k s.t. $2^k - 1 \geq N$.

$$\begin{aligned} \text{then } P\left(\bigcup_{n=N}^{\infty} A_n\right) &\geq P\left(\bigcup_{j=0}^{2^k-1} A_{2^k-1+j}\right) \leftarrow \{X_{2^k-1+j} = 1\} \\ &\Leftrightarrow u \in \left[\frac{j}{2^k}, \frac{j+1}{2^k}\right) \\ &= P\left(\bigcup_{j=0}^{2^k-1} \left\{u \in \left[\frac{j}{2^k}, \frac{j+1}{2^k}\right)\right\}\right) \\ &= P(u \in [0, 1)) = 1 \end{aligned}$$

$$\therefore P\left(\bigcup_{n=N}^{\infty} A_n\right) = 1 \quad \forall N$$

$$\therefore P(A_n \text{ i.o.}) = \lim_{N \rightarrow \infty} P\left(\bigcup_{n=N}^{\infty} A_n\right) = 1$$

$$\therefore P(X_n = 1 \text{ i.o.})$$

and $X_n \xrightarrow{a.s.} 0$

□

介绍一个可以给 almost sure convergence 转化的 theorem.

Thm: If $z_1, z_2, z_3, \dots$ are r.v., then T.F.A.E.
(the following are equivalent)

$$\begin{cases} (i) z_n \xrightarrow{a.s.} z \\ (ii) \forall \varepsilon > 0, P(|z_n - z| \geq \varepsilon \text{ i.o.}) = 0 \end{cases}$$

Proof: Case 1. (i) $\Rightarrow$ (ii)

思想: we prove by the way $\text{not}(ii) \Rightarrow \text{not}(i)$.

Suppose (ii) is not true,
then $\exists \varepsilon > 0$ s.t. $P(|z_n - z| \geq \varepsilon \text{ i.o.}) > 0$

$$\{|z_n - z| \geq \varepsilon \text{ i.o.}\} \Rightarrow \{z_n \not\rightarrow z\}$$

$$\therefore P(z_n \not\rightarrow z) \geq P(|z_n - z| \geq \varepsilon \text{ i.o.}) > 0$$

$$\Rightarrow P(z_n \rightarrow z) < 1 \Rightarrow z_n \not\xrightarrow{a.s.} z \quad \square$$

Case 2. (ii) $\Rightarrow$ (i)

$$\begin{aligned} P(z_n \rightarrow z) &= 1 - P(z_n \not\rightarrow z) \\ &= 1 - P(\exists \varepsilon > 0 \text{ s.t. } \forall N, \exists n \geq N \text{ s.t. } |z_n - z| \geq \varepsilon) \\ &= 1 - P\left(\bigcup_{\varepsilon > 0} \{\forall N, \exists n \geq N \text{ s.t. } |z_n - z| \geq \varepsilon\}\right) \\ &= 1 - P\left(\bigcup_{\varepsilon > 0} \{|z_n - z| \geq \varepsilon \text{ i.o.}\}\right) \end{aligned}$$

A big statement here for proving (using countable^{sub} additivity)

$$\bigcup_{\varepsilon > 0} \{|z_n - z| \geq \varepsilon \text{ i.o.}\} = \bigcup_{\substack{\varepsilon > 0 \\ \varepsilon \text{ rational}}} \{|z_n - z| \geq \varepsilon' \text{ i.o.}\}$$

continues
proving

Prove of the big statement suppose $A_k = \{|z_n - z| \geq k \text{ i.o.}\}$

$$\textcircled{1} \bigcup_{\varepsilon > 0} A_\varepsilon \supseteq \bigcup_{\substack{\varepsilon > 0 \\ \varepsilon' \text{ rational}}} A_{\varepsilon'} \text{ is obvious}$$

$$\textcircled{2} \bigcup_{\varepsilon > 0} A_\varepsilon \subseteq \bigcup_{\substack{\varepsilon' > 0 \\ \varepsilon' \text{ rational}}} A_{\varepsilon'}$$

pf: $\forall \varepsilon > 0, \exists$ a rational $\varepsilon' [0 < \varepsilon' < \varepsilon]$ or $\varepsilon' \in (0, \varepsilon)$.

$$\text{then } \{|z_n - z| \geq \varepsilon \text{ i.o.}\} \subseteq \{|z_n - z| \geq \varepsilon' \text{ i.o.}\}$$

$$\forall \varepsilon > 0, \exists \varepsilon' \text{ s.t. } A_\varepsilon \subseteq A_{\varepsilon'} \Rightarrow \bigcup_{\varepsilon > 0} A_\varepsilon \subseteq \bigcup_{\substack{\varepsilon' > 0 \\ \varepsilon' \text{ rational}}} A_{\varepsilon'}$$

From last part, we know that

$$P(\overline{Z_n \rightarrow z}) = 1 - P(\bigcup_{\varepsilon > 0} \{|z_n - z| \geq \varepsilon \text{ i.o.}\})$$

$$\begin{aligned} \text{by the } \nearrow &= 1 - P(\bigcup_{\substack{\varepsilon' > 0 \\ \varepsilon' \text{ rational}}} \{|z_n - z| \geq \varepsilon' \text{ i.o.}\}) \\ \text{"big statements"} & \end{aligned}$$

$$\begin{aligned} \text{by the } \nearrow &\geq 1 - \sum_{\substack{\varepsilon' > 0 \\ \varepsilon' \text{ rational}}} P(\{|z_n - z| \geq \varepsilon' \text{ i.o.}\}) \\ \text{countable subadditivity} & \\ &= 1 - \sum_{\varepsilon' > 0} 0 = 1 \end{aligned}$$

$$\therefore P(Z_n \rightarrow z) = 1 \quad \text{i.e. } Z_n \xrightarrow{\text{a.s.}} z$$

□

Cor: For r.v. $z_1, z_2, z_3, \dots$,
if $\forall \epsilon > 0, \sum_{n=1}^{\infty} P(|z_n - z| \geq \epsilon) < \infty$,

then, $z_n \xrightarrow{\text{a.s.}} z$

Easy Proof:

By Borel-Cantelli,

$$\sum_{n=1}^{\infty} P(|z_n - z| \geq \epsilon) < \infty \Rightarrow P(\{z_n - z \geq \epsilon\} \text{ i.o.}) = 0$$

$$\therefore \forall \epsilon > 0, P(|z_n - z| \geq \epsilon \text{ i.o.}) = 0$$

By the $\Rightarrow$ last theorem $\implies z_n \xrightarrow{\text{a.s.}} z$

• Strong Law of Large Numbers (SLLN)

If $X_1, X_2, \dots$ indep r.v., each with same finite mean m ,
and each with $E((X_i - m)^4) \leq q < \infty$, then if

$$M_n = \frac{1}{n}(X_1 + X_2 + \dots + X_n)$$

$$M_n \xrightarrow{\text{a.s.}} m$$

• Fact

if $\{X_n\}$ i.i.d., then don't need $E((X_i - m)^4) < \infty$